

SEASnet Setup Guide

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Account, VPN, SSH, file transfer, and VS Code for new UCLA CS/CSE students (CS 31, 32, 33, 35L), with separate guides for Windows, macOS, and Linux

This guide takes you from zero to a working connection to the SEASnet Linux servers, where most CS coursework is built and tested. Sections 1 to 3 apply to everyone. Then jump to the guide for your operating system. The last two sections cover troubleshooting and a cheat sheet.

What you need before you start

A UCLA Logon ID with multi-factor authentication (MFA) enrolled, enrollment in an engineering course (or an engineering major) with fees paid, and a laptop.

Contents

1	Activate your SEASnet account and set your password	Everyone
2	Connect from off campus with the Campus VPN	Everyone
3	Hosts, etiquette, and your disk quota	Everyone
4	Windows guide: PuTTY, WinSCP, PowerShell, VS Code	Windows
5	macOS guide: Terminal, SFTP, VS Code	macOS
6	Linux guide: OpenSSH, SFTP, rsync, VS Code	Linux
7	Troubleshooting	Everyone
8	Cheat sheet and sources	Everyone

Two different accounts

Your **UCLA Logon ID** (with MFA) is your campus identity and is what the VPN uses. Your **SEASnet account** is a separate lab account with its own username and password. It is what you type when you connect to the Linux servers. Mixing these up is the most common first-week mistake.

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1. Activate your SEASnet account

Who is eligible

SEASnet offers student accounts to registered engineering students, and to non-engineering students who are officially enrolled in an engineering course. In both cases your registration fees must be current.

Apply

- 1 Go to my.seas.ucla.edu and log in using the login button at the top right.
- 2 In the left menu, click **SEASnet Account**, then **Student Account Application**.
- 3 Wait 15 minutes before trying to use your new account.

Change your password or recover your username

- 1 Go to my.engineering.ucla.edu and log in at the top right.
- 2 In the **My Profile** block, click **Change SEASnet Password**.
- 3 Wait 15 minutes before using the new password.

The same steps show your SEASnet username if you forget it. SEASnet's pages refer to both the `my.seas.ucla.edu` and `my.engineering.ucla.edu` portals, so if one does not load, try the other.

Keep your account active

Your account is deactivated if your registration fees are not paid, or if you are no longer officially enrolled in any engineering class. An account that stays deactivated for over a year can be permanently deleted, so back up anything you care about to your own machine or a private Git repository.

2. Connect from off campus with the Campus VPN

The SEASnet Linux servers, SSH, and SFTP only work from the UCLA network. On campus Wi-Fi you are already on it. From home or a coffee shop, connect to the Campus VPN first.

- 1 Enroll in MFA at accounts.iam.ucla.edu/mfa. The VPN will not let you in without it.
- 2 Download the Cisco AnyConnect / Secure Client for your operating system from UCLA Digital & Technology Solutions: dts.ucla.edu [VPN page](#). That page links a setup article for Windows, macOS, iOS, Android, Chromebook, and Linux.
- 3 Connect to the campus VPN server, `vpn.ucla.edu`, and log in with your UCLA Logon ID (this address is from SEASnet's wireless page). If UCLA IT's setup article for your operating system gives different settings, follow the article.
- 4 Approve the MFA prompt, and keep the VPN connected for as long as your SSH or SFTP session is open.

VPN help: UCLA IT Support Center, (310) 267-HELP (4357) or help@it.ucla.edu. The IT Support Center does not provide technical support for the Linux VPN client.

3. Hosts, etiquette, and your disk quota

Which host to use

Host	Use it for
<code>lnxsrvt.seas.ucla.edu</code>	CS and other non-EE classes. SEASnet recommends this name because it balances load and simplifies maintenance. You must be enrolled in a class that is allowed on these servers.
<code>ugrad.seas.ucla.edu</code>	The general undergraduate host in SEASnet's SSH documentation. Hostnames depend on account type (fac, grad, ugrad, staff).
<code>lnxsrvt09.seas.ucla.edu</code> (a specific machine)	Only if your instructor or TA tells you to use it.

If your course syllabus or TA names a host, use that one. Every guide below uses `lnxsrvt.seas.ucla.edu` and `yourusername` as placeholders, so swap in your own username and your course's host.

Etiquette on shared servers

`lnxsrvt` is shared by many students. Keep heavy jobs short, do not leave runaway processes behind, and log out with `exit` when finished.

Home directory quota

Each student's home directory is currently limited to **3 GB**. After logging in, check your usage with `showdq`. If you go over the hard limit you may see no obvious error, but you will not be able to save files. SEASnet emails a warning when you pass the soft limit. For more space, email help@seas.ucla.edu.

4. Windows guide

Two ways to connect (PowerShell or PuTTY) and two ways to move files (WinSCP or scp). Pick one of each, or mix them.

4.1 Log in: option A, PuTTY (SEASnet's documented client)

PuTTY is a small SSH client that needs no installation. SEASnet's own PuTTY page walks through it.

- 1 Download `putty.exe` from the official PuTTY site linked on SEASnet's PuTTY page (chiark.greenend.org.uk/~sgtatham/putty). Save it to your desktop. Only download from the official site, since lookalike sites exist.
- 2 Double-click `putty.exe`. In the **Session** category, set **Host Name** to `lnxsrvc.seas.ucla.edu`, **Port** to 22, and **Connection type** to **SSH**.
- 3 Type a name such as `lnxsrvc` under **Saved Sessions** and click **Save**. Next time, select that name, then **Load** and **Open**.
- 4 Click **Open**. On the first connection PuTTY shows a security alert about the server's host key. Click **Accept** if you are connecting to the right host.
- 5 At the `login as:` prompt, type your SEASnet username and press Enter. Then type your password. Nothing appears on screen while you type it, which is normal.

Optional: change the font under **Window > Appearance** (SEASnet suggests Courier New, 10 point), then return to **Session** and **Save** again.

4.2 Log in: option B, PowerShell or Windows Terminal

Recent versions of Windows include an OpenSSH client, so no download is needed. Open Windows Terminal or PowerShell and run:

```
ssh yourusername@lnxsrvc.seas.ucla.edu
```

Answer yes to the host-key question the first time, then enter your password.

4.3 Transfer files: option A, WinSCP (graphical)

WinSCP is the free SFTP/SCP client that SEASnet lists for Windows. It shows your computer on the left and the server on the right, and you drag files between them. Download it from winscp.net/eng/docs/installation.

- 1 Open WinSCP. In the **Login** dialog choose **New Site**.
- 2 Set **File protocol** to **SFTP**, **Host name** to `lnxsrvc.seas.ucla.edu`, **Port number** to 22, and enter your SEASnet **User name** and **Password**.
- 3 Click **Save** so the site is stored, then click **Login**. Accept the host-key prompt the first time.
- 4 Your home directory appears in the right pane. Drag files or folders between panes to upload or download, or right-click a file to edit or delete it.

- 5 If you already saved a PuTTY session, WinSCP can import it from **Tools > Import Sites**.

Line endings

Files edited on Windows can carry Windows line endings, which break shell scripts on Linux with errors such as bad interpreter or ^M. In VS Code, click **CRLF** in the status bar and switch it to **LF** before you upload. On the server, `sed -i 's/\r$//' script.sh` also strips them.

4.4 Transfer files: option B, scp or PSFTP

Windows' built-in OpenSSH also provides `scp` and `sftp`. PuTTY's download page offers PSFTP, a command-line SFTP tool with no graphical interface.

```
scp hello.cpp yourusername@lnxsrv.seas.ucla.edu:~/
scp -r yourusername@lnxsrv.seas.ucla.edu:~/project .\project
```

4.5 Graphical Linux programs (X11)

Only needed if a course asks for it. Install and start an X server such as Xming, then in PuTTY open the **SSH > X11** settings, check **Enable X11 forwarding**, and save the session. In PowerShell, use `ssh -X yourusername@lnxsrv.seas.ucla.edu`.

4.6 VS Code with Remote-SSH (unofficial)

Read this first

SEASnet does not document or support Remote-SSH. It works for many students, but if it is slow, disconnects, or is blocked, fall back to PuTTY plus WinSCP, which are the documented tools. Check your course policy too.

- 1 Install [Visual Studio Code](#) and the **Remote - SSH** extension (published by Microsoft). Remote-SSH uses the Windows OpenSSH client, so make sure `ssh` works in PowerShell first (section 4.2).
- 2 If you are off campus, connect to the Campus VPN.
- 3 Press `Ctrl+Shift+P`, run **Remote-SSH: Connect to Host...**, and enter `yourusername@lnxsrv.seas.ucla.edu`.
- 4 Enter your SEASnet password when asked. The first connection installs a small VS Code server in your home directory.
- 5 Use **File > Open Folder** to open a folder on the server. The integrated terminal now runs on lnxsrv.

For a short name, add this to `C:\Users\YourWindowsName\.ssh\config` (create it if missing):

```
Host lnxsrv
  HostName lnxsrv.seas.ucla.edu
  User yourusername
```

The `.vscode-server` folder VS Code creates in your home directory counts against your 3 GB quota. If you hit the limit, run `showdq`, then delete big files or that folder. VS Code reinstalls what it needs next time.

5. macOS guide

macOS includes OpenSSH, so Terminal is all you need for connecting and copying files.

5.1 Log in

- 1 Open **Terminal** (Applications > Utilities > Terminal, or search with Spotlight).
- 2 Connect to the Campus VPN first if you are off campus (Section 2).
- 3 Run the command below. Answer yes to the host-key question the first time, then enter your SEASnet password. Nothing shows while you type it.

```
ssh yourusername@lnxsrv.seas.ucla.edu

# equivalent form
ssh lnxsrv.seas.ucla.edu -l yourusername
```

Type `exit` to log out.

5.2 Transfer files

Run these in Terminal on your Mac, not while logged in to the server.

```
# upload a file to your home directory
scp hello.cpp yourusername@lnxsrv.seas.ucla.edu:~/

# download a folder
scp -r yourusername@lnxsrv.seas.ucla.edu:~/project ~/Desktop/project

# interactive session
sftp yourusername@lnxsrv.seas.ucla.edu
sftp> ls
sftp> put hello.cpp
sftp> get results.txt
sftp> quit
```

If you prefer a graphical file browser, any SFTP client works with the same host, username, port 22 and your SEASnet password. Cyberduck is a common free one. SEASnet's documentation only covers the command-line tools above.

5.3 Graphical Linux programs (X11)

Only needed if a course tells you to run a graphical program. macOS needs an X server first: install XQuartz, then log out and back in to your Mac. Connect with:

```
ssh -X yourusername@lnxsrv.seas.ucla.edu
```

5.4 VS Code with Remote-SSH (unofficial)

Read this first

SEASnet does not document or support Remote-SSH. If it is slow, disconnects, or is blocked, fall back to Terminal ssh plus scp. Check your course policy too.

- 1 Install [Visual Studio Code](#) and the **Remote - SSH** extension.
- 2 If you are off campus, connect to the Campus VPN.
- 3 Press Cmd+Shift+P, run **Remote-SSH: Connect to Host...**, and enter `yourusername@lnxsrv.seas.ucla.edu`.
- 4 Enter your SEASnet password when asked. The first connection installs a small VS Code server in your home directory.
- 5 Use **File > Open Folder** to open a folder on the server. The integrated terminal now runs on lnxsrv.

For a short name, add this to `~/.ssh/config` on your Mac (create the file if it does not exist):

```
Host lnxsrv
  HostName lnxsrv.seas.ucla.edu
  User yourusername
```

After that, `ssh lnxsrv` works in Terminal and `lnxsrv` appears in VS Code's host list. The `.vscode-server` folder counts against your 3 GB quota, so run `showdq` if you get low on space.

6. Linux guide

Almost every distribution ships OpenSSH. If yours does not, install the client with your package manager.

6.1 Install the client if needed

```
# Debian / Ubuntu
sudo apt install openssh-client

# Fedora
sudo dnf install openssh-clients
```

6.2 Connect the VPN and log in

Off campus, you need the Campus VPN first. UCLA IT publishes a Linux client article on the DTS VPN page (Section 2), but the IT Support Center does not provide technical support for it, so budget extra time. Then:

```
ssh yourusername@lnxsrv.seas.ucla.edu
```

Answer yes to the first host-key question, then enter your SEASnet password.

6.3 Transfer files

```
# upload / download
scp hello.cpp yourusername@lnxsrv.seas.ucla.edu:~/
scp -r yourusername@lnxsrv.seas.ucla.edu:~/project ./project

# interactive
sftp yourusername@lnxsrv.seas.ucla.edu

# sync a folder (only changed files are sent)
rsync -avz ./project/ yourusername@lnxsrv.seas.ucla.edu:~/project/
```

Mind the trailing slashes in `rsync`. With a slash after the source folder, the folder's contents are copied into the destination. Without it, the folder itself is copied inside the destination.

Optional, community technique: mount your SEASnet home directory as a local folder with `sshfs` (install the `sshfs` package first). This is not documented by SEASnet.

```
mkdir -p ~/lnxsrv
sshfs yourusername@lnxsrv.seas.ucla.edu: ~/lnxsrv

# unmount when done
fusermount -u ~/lnxsrv
```

6.4 Graphical Linux programs (X11)

Use the `-X` flag. This is the exact example on SEASnet's Linux Servers page. It works out of the box on X11 sessions, and Wayland desktops usually handle it through XWayland.

```
ssh -X lnxsrv.seas.ucla.edu -l yourusername
```

6.5 VS Code with Remote-SSH (unofficial)

Read this first

SEASnet does not document or support Remote-SSH. If it is slow, disconnects, or is blocked, fall back to plain ssh plus scp or rsync. Check your course policy too.

- 1 Install [Visual Studio Code](#) and the **Remote - SSH** extension.
- 2 If you are off campus, connect the Campus VPN.
- 3 Press Ctrl+Shift+P, run **Remote-SSH: Connect to Host...**, and enter `yourusername@lnxsrv.seas.ucla.edu`.
- 4 Enter your SEASnet password when asked, then use **File > Open Folder** to open a folder on the server.

For a short name, add this to `~/.ssh/config`:

```
Host lnxsrv
  HostName lnxsrv.seas.ucla.edu
  User yourusername
```

The `.vscode-server` folder counts against your 3 GB quota, so run `showdq` if you get low on space.

7. Troubleshooting

Symptom	Likely cause and fix
Connection timed out, or no route to host	You are not on the campus network. Connect the Campus VPN (Section 2) and retry.
Permission denied (password)	Check that you are using your SEASnet username and password, not your UCLA Logon ID. If you just applied or changed your password, wait the full 15 minutes.
Login worked before, now fails	Your account may be deactivated because fees are unpaid or you are not enrolled in an engineering class. Check your enrollment and fees, then contact SEASnet.
Cannot save files or odd errors while editing	You may be over quota. Run <code>showdq</code> and delete large or old files.
VPN will not connect	Confirm MFA enrollment and that you are using the client from UCLA DTS. Contact the IT Support Center at (310) 267-HELP.
PuTTY or WinSCP closes right after login	Check the username and password first. Then confirm you are on the VPN. Try connecting with PowerShell <code>ssh</code> to see the exact error message.
Script fails with bad interpreter or ^M	The file has Windows line endings. See the line-endings note in Section 4.3.
Host key warning after a server change	Do not ignore it blindly. Ask your TA or SEASnet whether the server was changed before you edit <code>~/.ssh/known_hosts</code> .
Still stuck	Email help@seas.ucla.edu with your username (never your password), the exact error text, and what you already tried.

8. Cheat sheet and sources

Settings for graphical clients (PuTTY, WinSCP, Cyberduck)

Field	Value
Host name / server	<code>lnxsrv.seas.ucla.edu</code> (or your course's host)
Port	22
Protocol	SSH for PuTTY, SFTP for WinSCP and other file clients
Username / password	Your SEASnet username and password
Off campus	Campus VPN connected first

Terminal commands (macOS, Linux, Windows PowerShell)

Task	Command
Log in	<code>ssh yourusername@lnxsrv.seas.ucla.edu</code>
Log in with graphics	<code>ssh -X yourusername@lnxsrv.seas.ucla.edu</code>
Upload a file	<code>scp file.cpp yourusername@lnxsrv.seas.ucla.edu:~/</code>
Download a folder	<code>scp -r yourusername@lnxsrv.seas.ucla.edu:~/dir ./dir</code>
Interactive transfer	<code>sftp yourusername@lnxsrv.seas.ucla.edu</code>
Check disk quota (on the server)	<code>showdq</code>
Log out	<code>exit</code>

New to the shell? SEASnet has a short guide: [Basic Unix commands](#).

Sources (checked September 29, 2026)

- SEASnet, Apply for Student Account: seasnet.ucla.edu/apply-for-student-account/
- SEASnet, Who is Eligible: seasnet.ucla.edu/seasnet-accounts/
- SEASnet, Changing Your Password: seasnet.ucla.edu/changing-password/
- SEASnet, Linux Servers and lnxsrv: seasnet.ucla.edu/linux-servers/ and seasnet.ucla.edu/lnxsrv/
- SEASnet, Unix Clients (SSH, SFTP): seasnet.ucla.edu/unix-clients/
- SEASnet, Secure Remote Login and File Transfer (PuTTY, WinSCP, OpenSSH): seasnet.ucla.edu/secure-remote-login-file-transfer/
- SEASnet, PuTTY: seasnet.ucla.edu/putty/
- SEASnet, Account Quotas: seasnet.ucla.edu/seasnet-account-quotas/
- SEASnet, Wireless Network (VPN server address): seasnet.ucla.edu/seasnet-wireless-network/
- UCLA DTS, VPN Clients and When to use the Campus VPN: dts.ucla.edu/products-services/software-downloads/virtual-private-network-vpn

The WinSCP steps, the macOS and Linux sections, the VS Code Remote-SSH sections, and the troubleshooting table are practical advice, not official SEASnet documentation. Server names, quotas, and VPN clients can change, so confirm against the sources above if something no longer matches.